

Education

Ph.D., Princeton University. Astrophysical Sciences. 2022 - Present
H.B.Sc., University of Toronto. Astronomy & Astrophysics and Statistics. 2018-2022

Publications

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3. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy. Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution, in prep.
 2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey. 2022, ApJ, 925, 1. [2111.09327](#).
 1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. [2105.11572](#).

Research Experience

2021 May - Present	Predicting the ages of stars with machine learning Supervisors: Dr. Joshua Speagle , Dr. Neige Frankel , Dr. J. Ted Mackereth Independent research. Probabilistic predictions of stellar ages using normalizing flows. Quantifying the contribution of stellar/Galactic information to age estimates.
2021 Sep - 2022 Jun	Narrowband upgrade for the Dragonfly Telephoto Array Supervisor: Prof. Roberto Abraham Senior thesis. Containerized camera control software in C++ and Rust with real-time, multi-threaded data processing.
2021 Jul - 2022 Jun	Constraining the strength of thermal tides with Stan Supervisor: Prof. Norman Murray Independent research. Fitting dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological time.
2020 Aug - 2021 Nov	Estimating the mass of the Galaxy with Bayesian hierarchical models Supervisors: Prof. Gwendolyn Eadie , Prof. Norman Murray , Dennis Zaritsky , Dr. Joshua Speagle Undergraduate fellowship with the Canadian Institute for Theoretical Astrophysics (CITA). Developed a hierarchical Bayesian model for modelling the mass of the Galaxy using a fast and scalable sampling algorithm. Collaborated with the H3 Survey team and presented results in ApJ.
2020 May - 2021 May	Molecular gas in high-redshift galaxies with ALMA Supervisor: Prof. Allison Man Summer undergraduate research program with the Dunlap Institute. Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ.

Honours and Awards

2022 Jan	Astrostatistics Student Paper Competition Finalist Astrostatistics Interest Group of the American Statistical Association
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2021 - 2022	Dean's List Scholar University of Toronto
2021 May - 2021 Aug	Undergraduate Summer Research Fellowship, (\$8,396 CAD) Canadian Institute for Theoretical Astrophysics, University of Toronto
2020 May - 2020 Aug	Summer Undergraduate Research Program Award (\$9,500 CAD) Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Talks

2022 August	JSM 2022, Washington, D.C. Contributed talk: "The Mass of the Milky Way from the H3 Survey."
2021 August	SDSS 2021 Collaboration Meeting, JHU/Online Lightning talk: "Predicting the ages of stars with machine learning."
2021 May	Stellar Stats Workshop, UofT/Online Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models."
2021 May	Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online Lightning talk: "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$."
2021 Apr	Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online "Molecular gas in gravitationally lensed galaxies at $z = 2.9$."
2020 Oct	REQUIEM Galaxies Telecon, Online "Molecular gas in gravitationally lensed galaxies at $z = 2.9$."

Posters

2021 May	Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$."
2020 Aug	Summer Undergraduate Research Program Poster Session, UofT/Online "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$." Awarded prize for one of top three posters.

Outreach

2019 - 2021	Space Science Journalist (Current in Space segment), <i>The Star Spot</i> podcast Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.
2015 - 2018	Python Workshop Lead, Sir Winston Churchill Secondary, Vancouver, B.C. Coordinated and delivered workshops to prepare students for national programming contest.

Media

2021 Jul	National Radio Astronomy Organization (NRAO) eNews Digital newsletter feature: "Molecular gas in high redshift galaxies". https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas
2020 Jul	SURP Student of the Week Feature Interview: https://www.dunlap.utoronto.ca/surp-sow/sow12020/

Technical Skills

JAX, Stan, Pytorch, SQL, R, C++, Rust, BLAS/LAPACK, L^AT_EX, regex, bash, git, Docker, Javascript, React